

L12: Generative Deep Learning

CSci 560 Deep Learning w/ Python (Chollet) Ch. 12

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L12.1 Text Generation

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How do you Generate Sequence Data?

- Train a model (usually Transformer or RNN) to predict next token
 - e.g. like the encoder / decoder architecture, though we only have 1 sequence so only need decoder
- Any network that can model the probability of the next token given the previous ones is called a **language model**.
- A language model captures the **latent space** of language: its statistical structure.
- **Sample** from a language model: feed it an initial string (**conditioning data**) ask it to generate next token.
 - Add the generated output back to the input data, repeat

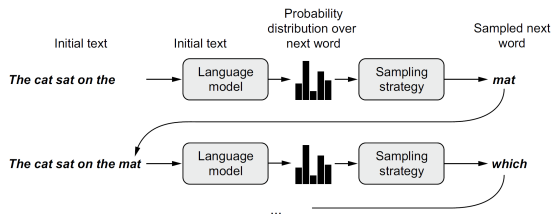


Figure 1: The process of word-by-word text generation using a language model. (Chollet, 2021, pg.367)

The Importance of the Sampling Strategy

- **Greedy sampling** consists of always choosing the most likely next token.
- **Stochastic sampling** sampling from the probability distribution for the next character.
- Useful to **control the amount of randomness** in the sampling process **softmax temperature**

```
def reweight_distribution(original_distribution, temperature=0.5):  
    """Given a softmax probability distribution, reweight the probabilities  
    using a temperature variable. Temperature 0 = greedy sampling, temperature  
    1 = uniform like (completely random) distribution  
  
    Parameters  
    -----  
    original_distribution : ndarray  
        Vector of probability over a vocabulary, probability must sum to 1.0  
    temperature : float  
        Factor quantifying the entropy of the output distribution  
  
    Returns  
    -----  
    new_distribution : ndarray  
        A reweighted version of the original distribution.  
    """  
    distribution = np.log(original_distribution) / temperature  
    distribution = np.exp(distribution)  
    # sum may longer be 1.0, so divide by the sum to obtain new probability  
    # distribution that sums to 1.0  
    return distribution / np.sum(distribution)
```



Summary

- You can generate discrete sequence data by training a model to predict the next token(s), given previous tokens.
- In the case of text, such a model is called a **language model**. It can be based on either words or characters.
- Sampling the next token requires a balance between adhering to what the model judges likely, and introducing randomness.
- One way to handle this is the notion of softmax temperature. Always experiment with different temperatures to find the right one.
- Don't expect to generate meaningful text (even for a chat GPT level of training data), all you are doing is sampling data from a statistical model of which words come after which words. Language models are all form and no substance.



L12.2 DeepDream

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DeepDream

- DeepDream is an artistic image-modification technique that uses the representations learned by convolutional neural networks.
- The DeepDream algorithm is almost identical to the convnet filter-visualization technique introduced previously consisting of running a convnet in reverse:
 - doing gradient ascent on the input to the convnet in order to maximize the activation of a specific filter in an upper layer of the convnet.

DeepDream uses this same idea, with a few simple differences:

- With DeepDream, you try to maximize the activation of entire layers rather than that of a specific filter, thus mixing together visualizations of large numbers of features at once.
- You start not from blank, slightly noisy input, but rather from an existing image—thus the resulting effects latch on to preexisting visual patterns, distorting elements of the image in a somewhat artistic fashion.
- The input images are processed at different scales (called octaves), which improves the quality of the visualizations.

L12.3 Neural Style Transfer

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L12.4 Generating Images with Variational Autoencoders

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L12.5 Introduction to Generative Adversarial Networks

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Chollet, F. (2021). *Deep learning with python* (second). Manning.